Status	Finished
Started	Thursday, 29 October 2026, 7:46 PM
Completed	Thursday, 29 October 2026, 7:50 PM
Duration	3 mins 15 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search

b. Linear search ✓

c. Both (1) & (3)

d. Hash search

Question **2**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500

b. 10 ✓

c. 100

d. 1000

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned

b. An error is raised

c. The last element is returned

d. The search is considered unsuccessful ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By comparing each element in the list sequentially ✓
c. By dividing the list into halves
d. By sorting the list first

Question **5**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Both (1) & (3)
c. Linear
d. Hash

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Searching ✓
c. Discovering
d. Mining

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. There must be a mechanism to access middle element directly
d. Must use a sorted array

Question **8**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search works on unsorted lists
c. Binary search does not require dividing the list
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O(1) ✓
d. O👎

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. n ✓
d. log n

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94
b. 89 and 99
c. 90 and 99 ✓
d. 89 and 94

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the left sublist ✓
d. In the entire list

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 90 and 100
c. 94 and 99
d. 89 and 94

Question **15**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. When the item is somewhere in the middle of the array ✓
c. Item is the last element in the array or item is not there at all
d. When the item is the last element in the array